

MarEng Radio Communication

GLOBAL MARITIME DISTRESS AND SAFETY SYSTEM (GMDSS) DIGITAL SELECTIVE CALL

GENERAL



The GMDSS enables Maritime Rescue Centres and radio stations to keep automatic watch on the distress channels. A ship equipped with a DSC apparatus can make a distress call by simply pressing a button, which results in an automatic alarm signal at the maritime rescue centre nearest to the vessel.

DSC can be used for all kinds of radio calls between ship-ship and ship-shore-ship. DSC calls are made on channel 70, which means that any voice communication on this channel is not allowed.

The DSC call includes and shows the following information:

- Format or Type of Call
- Receiver or Party ID
- Call Category or Priority
- Telecommand
(follow-up radio communication)
- Working Channel

1. FORMAT OR TYPE OF CALL

One of the following are chosen on the DSC apparatus:

- | | |
|----------------------------------|--|
| - distress alert / distress call | in a distress |
| - all ships | a call to all stations |
| - selective/individual | a call to one station
(e.g. in routine calls) |
| - group | a group of stations |
| - area / geographical call | to stations in a
particular area |

2. PARTY/RECEIVER ID (IDENTIFICATION)

The receiver of the call (Party ID) you can click either the Maritime Mobile Service Identity (MMSI) number of the receiving station, a group call number or area call number. The MMSI always contains nine digits.

Example of MMSI Number system:

MMSI Vessel 230252000	Explanation 230 country code (Finland) always three digits 252000 (ship's number)
MMSI Coast Station 002301000	Explanation 00 means coast station in general 230 country code (Finland) number of particular coast station 1000 (Turku Maritime Rescue Coordination Centre)

3. CALL CATEGORY OR PRIORITY

- distress
- urgency
- safety

Routine Communication can be of two kinds:

- routine
- ship's business

4. TELECOMMAND

This gives information on what kind of follow-up radio communication the sender wishes to have. It also gives information on what kind of message the sender is giving. The most common telecommand choices include:

- telephony G3E/F3E simplex
 - o voice communication on VHF
- distress acknowledgement
 - o request for acknowledgement of distress call
- distress relay alert
 - o request for relay of distress
- ship's position
 - o request for a vessel's position to be shown on DSC (routine call)

5. WORKING CHANNEL

The one who sends a DSC call can suggest a working channel for continued voice radio communication.

NOTE: When a vessel is calling a coast station it does not suggest a working channel. The coast station announces which radio channel to continue on.

VHF DSC CONTROLLER

DSC Status: Ch 70

Position: 61°14'N 020°53'E at 16:49 UTC

1 ABC	2 DEF	3 GHI
4 JKL	5 MNO	6 PQR
7 STU	8 VWX	9 YZ
*	0	#

Supply
On/Off

DISTRESS

◀

▶

Cancel

▶

◀

Enter

Main
menu

Call

ROUTINE COMMUNICATION WITH DSC CALL

EXAMPLE 1

A RADIO MESSAGE FOR SUOMENLINNA COAST GUARD STATION

COMPARE THE FOLLOWING TO THE EXAMPLES SENT WITHOUT DSC:

A Russian yacht Marlene is approaching the Finnish coast on its way to spend a holiday partly in Helsinki and partly by sailing in the Finnish archipelago. According to European union laws a foreign pleasure boat must report for passport control when she has entered the territorial waters of one of the member states. The Marlene is a luxury yacht and is linked to GMDSS and has DSC. It is 12.30 at Suomenlinna Coast Guard Station on a fine summer day.

The DSC at Suomenlinna starts signalling and shows the following message:

Format:	SELECTIVE
Party ID:	002300005
Category	ROUTINE
Telecom 1	TELEPHONY (F3E/G3E SIMPLEX)
Work channel	NONE
Transmitted	CH 70

NOTE: Marlene does not suggest a working channel. In communication between ship and coast station, it is always the coast station that determines the choice of working channel.

SELECTIVE	directed to a specific station only
PARTY ID	002300005 the MMSI number of Suomenlinna Coast Guard Station

The Coast Guard station sends the following acknowledgement:

Format: SELECTIVE
Party ID: 273678951
Category: ROUTINE
Telecom 1: TELEPHONY (F3E/G3E SIMPLEX)
Work Channel: 74
Transmitted: Ch 70

Party ID 273678951 = the MMSI number of Marlene
Format SELECTIVE = directed towards one station

The Coast Guard Station suggests:

Work channel 74

After this Marlene CONTINUES BY VOICE COMMUNICATION on channel 74:

*Calling Finnish Coast Guard Suomenlinna, Finnish Coast Guard Suomenlinna,
This is Russian vessel Marlene, JAAA,
Russian vessel Marlene JAAA,
On VHF channel 74.
Over.*

*Russian Vessel Marlene JAAA,
This is Finnish Coast Guard Station
Suomenlinna,
Over.*

*Finnish Coast Guard station,
This is Russian yacht Marlene,
My position is: Latitude: 59° 56.9' N Longitude : 024° 56.3' E
I am in the approach channel passing Helsinki Lighthouse. I am proceeding to the Coast Guard
Station for passport control. REQUEST: Please give instructions to reach Suomenlinna Coast
Guard Station. Over.*

*Marlene,
This is Finnish Coast Guard.
INSTRUCTIONS: Proceed along the fairway. Pass
Harmaja Lighthouse and proceed along the fairway
distance 1.6 miles. Take fairway depth 9.6, course 328°.
Proceed distance 0.8 miles. You can see the information
on your starboard side.*

*I repeat: Proceed along the fairway. Pass Harmaja
Lighthouse and proceed along the fairway distance 1.6
miles. Take fairway depth 9.6, course 328°. Proceed
distance 0.8 miles. You can see the information on your
starboard side. Over.*

*Finnish Coast Guard ,
This is Marlene.*



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UNDERSTOOD. I will proceed along the fairway and pass Harmaja Lighthouse. I will take fairway depth 9.6, steering course 328°. I will proceed along the fairway distance 0.8 miles ahead. I can see information sign on starboard side. THANK YOU AND OUT.

AS YOU NOTICED THE DSC CALL REPLACES THE INITIAL VOICE COMMUNICATION CALL AND ANSWER ON CHANNEL 16. CHANGING TO A WORK CHANNEL ALSO HAPPENED VIA DSC.

THUS A TRADITIONAL VOICE TRANSMISSION ON CHANNEL 16 DIFFERS SLIGHTLY IN THE INITIAL STAGE BUT RADIO PROCEDURE IS OTHERWISE THE SAME.

DSC DISTRESS COMMUNICATION

DSC DISTRESS INFORMATION:

A distress alert by Digital selective call should include the following information:

- the nature of call is DISTRESS
- MMSI number of vessel in distress
- position of vessel in distress in coordinates
- position at TIME
- nature of distress
- suggestion for follow up communication (voice or text)

NOTE:

- A distress call is always addressed at all stations.
- Voice communication always takes place on channel 16.

NATURE OF DISTRESS

The DSC apparatus has a menu including the following choices for nature of distress:

- fire, explosion
- flooding
- collision
- grounding
- listing in danger of capsizing
- sinking
- disabled and adrift
- undesignated distress
- abandoning ship
- piracy/armed robbery attack
- man overboard
- EPIRB

EXAMPLE - DISTRESS CALL DSC

Area A1 on VHF

The DSC shows the following information:

Format:	DISTRESS
Distress ID	230136780 (MMSI number of vessel)
Position:	59 23 N 020 36 E
Pos.UTC:	1345 UTC
Nature of distress:	LISTING IN DANGER OF CAPSIZING
Telecom:	TELEPHONY (F3E/G3E SIMPLEX)
Transmitted:	Ch 70

EXAMPLE - DISTRESS ACKNOWLEDGEMENT DSC

Area A1 on VHF

The Distress call is acknowledged by MRCC Turku:

The acknowledgement repeats the information in the Distress Call.

It is directed to and seen on the DSC apparatus by all ships in the vicinity.

Format:	ALL SHIPS
From:	002301000
Category:	DISTRESS
Telecom 1:	DISTRESS ACKNOWLEDGEMENT
Distress ID:	230136780
Position:	59 23 N 020 36 E
Pos.UTC:	1345 UTC
Nature of distress:	LISTING IN DANGER OF CAPSIZING
Telecom:	TELEPHONY (F3E/G3E SIMPLEX)
Transmitted:	Ch 70

**AFTER THIS FOLLOWS VOICE COMMUNICATION ON CHANNEL 16
DISTRESS MESSAGE**

MAYDAY

This is 230136780

VERA

CALL SIGN JULIETT ALFA BRAVO CHARLIE

POSITION: 59° 23' NORTH 020° 36' EAST

Heavy leaking, danger of capsizing. Sea is rough.

Three men on board, five men in dinghy.

REQUEST: I require immediate assistance.

DISTRESS ACKNOWLEDGEMENT

MAYDAY

VERA VERA VERA

Call sign Juliett Alfa Bravo Charlie

This is RESCUE CENTRE TURKU RESCUE CENTRE TURKU

RESCUE CENTRE TURKU

RECEIVED MAYDAY

Over

**FOLLOWED BY A MAYDAY RELAY FROM THE RESCUE
CENTRE ON CHANNEL 16...**

MAYDAY RELAY MAYDAY RELAY MAYDAY RELAY



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ALL STATIONS ALL STATIONS ALL STATIONS

This is 002301000

Rescue Centre Turku Rescue Centre Turku Rescue Centre Turku

Following received from Vera Juliett Alfa Bravo Charlie at 1200 UTC on VHF channel 16 :

Begins

Mayday

This is Vera

Juliett Alfa Bravo Charlie

In position 59° 23' NORTH 020° 36' EAST

Heavy leaking, danger of capsizing. Sea is rough.

Three men on board, five men in dinghy.

REQUEST: I require immediate assistance.

Ends

Ships in area report to Rescue centre Turku

Over

DSC SAFETY COMMUNICATION

The following is an example of the MESSAGE STRUCTURE of a Safety Call with DSC by Turku Radio. This call is seen by vessels in the area.

Format:	All ships
From:	MMSI NUMBER (002300230)
Category	SAFETY
Telecom:	TELEPHONY (F3E/G3E SIMPLEX)
Work Channel	16
Transmitted:	Ch 70

It is followed by a safety call by voice communication sent by Turku Radio on VHF channel 16, so that every vessel, even smaller ones without DSC will receive the warning. The safety message is then read on the traffic channel:

SAFETY CALL ON CHANNEL 16:

SECURITÉ SECURITÉ SECURITÉ
ALL STATIONS ALL STATIONS ALL STATIONS
THIS IS: MMSI NUMBER
NAME OF STATION 1 X:
NAVIGATIONAL WARNING
PLEASE SWITCH TO TRAFFIC CHANNELS

SAFETY MESSAGE ON TRAFFIC CHANNEL:

SECURITÉ SECURITÉ SECURITÉ
ALL STATIONS ALL STATIONS ALL STATIONS
THIS IS: MMSI NUMBER
NAME OF STATION 1 X:
NAVIGATIONAL WARNING



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DSC URGENCY COMMUNICATION

A vessel or a coast station can send an urgency call (Pan-Pan call) by DSC. The DSC call contains information on which channel the voice communication will be read (Channel 16). The call can be directed to all stations or directly to the nearest Maritime Rescue Centre.

A Pan-Pan message may be sent in, for example, the following situations:

- a person has fallen overboard and the ship needs help with search and rescue
- signals have been observed that might be distress signals
- when reporting on missing vessels
- if a person on board a vessel needs immediate medical assistance
- red rockets have been observed

EXAMPLE

The following is an example of an Urgency call to all stations by DSC:

A

Format	ALL SHIPS
Category	URGENCY
Telecom 1	TELEPHONY (F3E/G3E SIMPLEX)
Work Channel	16
Transmitted	70

A vessel can send an urgency call directly to the nearest Maritime Rescue Centre. Example of a call sent to MRCC Turku:

B

Format	SELECTIVE
Party ID	002301000
Category	URGENCY
Telecom 1	TELEPHONY F3E/G3E SIMPLEX)
Work channel	NONE
Transmitted	CH 70

NOTE: when sending a DSC call to a Rescue Centre or Coast Station a vessel does not suggest the working channel. This is done by the Coast Station or Rescue Centre.